Smart Gym Management System: A Full-Stack, Secure, and Scalable Web Platform for Modern Fitness Centres

Aryan Suthar
B.Tech (Information Technology)
School of Computer Science, Engineering & Technology
ITM SLS Baroda University
aryan.suthar@itmbu.ac.in

Dr. Ashutosh Abhangi
Associate Professor
School of Computer Science, Engineering & Technology
ITM SLS Baroda University
ashutosh.cse@itmbu.ac.in

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Abstract

Smart GMS was architecturally designed to explicitly address the financial and operational barriers currently overwhelming smaller operators in the competitive gym software marketplace. Throughout our initial investigations, we observed that prevalent commercial solutions heavily burden independent facilities with exorbitant recurring subscription fees, rigid interface configurations, and an inability to export patron data or integrate diverse third-party APIs absent explicit vendor clearance. To resolve these operational constrictions, we engineered the Smart Gym Management System (Smart GMS), a responsive full-stack web application programmed natively via React 18, Spring Boot 3, and a thoroughly robust Oracle Database cluster. This research formally introduces our bespoke schema-level multi-tenant layout uniquely tailored to simultaneously service multiple separate gym locations utilizing identically shared infrastructure. Our platform operates via a uniquely structured four-tier Role-Based Access Control (RBAC) hierarchy, leveraging JSON Web Token (JWT) stateless validations and time-sensitive Two-Factor Authentication (2FA). We actively incorporated a fully integrated STOMP WebSocket communication plane and a custom CRON-triggered lifecycle coordinator specifically automating membership timelines. Empirical stress scenarios targeting our system, empirically measured in this study, confirm our REST endpoints effortlessly preserve 95th-percentile latencies beneath 200 milliseconds, whilst our chat messaging achieves seamless transmission velocities under 15 milliseconds. Concurrently, our codebase securely sustains an 87% testing coverage ceiling, validating production safety protocols. Our subsequent analytical breakdowns prove that Smart GMS overwhelmingly delivers an exhaustive feature suite thoroughly overshadowing prominent rivals, acting fundamentally as a self-hosted, open-source solution that eliminates recurring licensing fees entirely.

Keywords: Gym Administration Software, Spring Boot, React, Oracle Database, Full-Stack Architecture.

I. Introduction

The fitness software market exceeded USD 87 billion as of 2023, according to recent industry analysis [25], undergoing sweeping technological advancements. Conventional gym administration processes—often dependent on isolated software tools or manual record-keeping—frequently result in suboptimal user experiences. During the requirements phase, we observed that platforms such as Mindbody impose steep financial overhead for essential features. Furthermore, our analysis indicated these commercial applications severely limit third-party API configurations and enforce rigid operational pathways, substantially deterring smaller gym owners from migrating toward fully digital workflows.

To overcome these barriers, this study introduces the **Smart Gym Management System (Smart GMS)**. Designed as a fully open-source, full-stack web application, the platform streamlines critical workflows including membership tracking, trainer scheduling, instant messaging, and financial reporting within a highly secure multi-tenant environment.

The primary contributions of this research are:

- The conceptualization and deployment of a nested four-tier RBAC framework (Admin → Owner → Trainer → Member) ensuring strict permission inheritance.
- A dedicated real-time communication module built on WebSockets, demonstrably sustaining internal transmission bursts effectively.
- An automated, CRON-powered membership lifecycle controller that handles expiration detection, renewal processing, and asynchronous notifications.
- A time-decay weighting mechanism for trainer evaluations, ensuring that recent instructional performance disproportionately influences the overall rating.
- A detailed quantitative assessment comparing Smart GMS against five prominent commercial gym software providers across eight operational domains and pricing models.
- A fully open-source, royalty-free architectural blueprint rigorously validated against the OWASP Top 10 security framework.

The subsequent sections of this document are structured as follows: Section 2 examines existing literature and provides a market comparison; Section 3 outlines the underlying software ar-

chitecture; Section 4 covers formal system modeling; Section 5 details the functional capabilities; Section 6 defines the mathematical algorithms utilized; Section 7 discusses empirical performance metrics; Section 8 investigates security implementations; Section 9 identifies current system boundaries and limitations; Section 10 summarizes the findings; Section 11 proposes future enhancements; and Section 12 offers acknowledgments.

II. Literature Review

The physical fitness sector has seen a profound shift toward digital integration. Gym software has transitioned from basic entry-tracking utilities to all-encompassing ecosystems that manage client engagement, staff coordination, and revenue analytics. According to industry analysis [25], the global fitness software market exceeds USD 87 billion globally and is projected to expand significantly by 2030 (achieving a 6.2% CAGR), propelled heavily by emerging connected fitness solutions and digital health portals.

Proprietary software such as Zen Planner, Mindbody, and GymMaster grant gym owners powerful tools for payment processing, facility scheduling, and biometric tracking. However, these applications carry steep barriers to entry. Specifically, our analysis reveals these corporate SaaS tools mandate expensive monthly usage fees that marginalize smaller business owners, actively lock databases within closed architectures preventing third-party integrations, and dictate painfully rigid operational flows [6]. Further supporting this perspective, a 2022 industry survey found that nearly two-thirds of independent gym operators cited cost and complexity as direct barriers to digitisation [25].

Reviewing the economic structuring of modern fitness software reveals a deep reliance on the Software-as-a-Service (SaaS) paradigm. While this recurring billing mechanism generates consistent revenue for developers, our observations confirm it systematically penalizes single-location gyms that lack the sheer client volume necessary to justify premium enterprise-tier subscriptions. By contrast, deploying an open-source framework completely uncouples the initial software licensure expenses from ongoing hardware hosting fees. This strategic separation permits business owners to vigorously invest their scarce capital directly into stronger local network infrastructure instead of engaging in perpetual software rentals.

Current scholarly investigations have pinpointed the essential characteristics of successful fitness platforms. Sharma et al. [12] identified goal visualisation, gamified progression, social networking, personalised scheduling, and optimised push notifications as universally critical structural patterns accelerating consumer loyalty. The Smart GMS incorporates all five aspects organically through customized member dashboards, trainer rating leaderboards, a built-in instant messaging subsystem, a dedicated personal training (PT) scheduling interface, and an automated alert engine. Furthermore, research by Chen and Lee [13] demonstrated that direct digital trainer-client communication increased annual retention by 23%, a finding that directly motivated our WebSocket module's development.

Architectural studies have significantly influenced our infrastructure choices. Rodriguez [14] evaluated three specific multi-tenancy strategies and concluded that schema-level isolation best balances performance and data privacy for platforms serving under 500 tenants—a finding directly applicable to Smart GMS's targeted operational scale. In parallel context, Williams [15] provided field data from fitness tech deployments indicating that well-structured monolithic architectures with defined domain boundaries frequently outperform premature microservices implementations for user bases spanning up to 50,000 active individuals. This research definitively reinforced our choice to safely construct Smart GMS as a modular monolith. Security benchmarks regarding PCI-DSS guidelines and rigorous penetration recommendations detailed by OWASP [11] systematically shielded our progressive iteration blocks.

To quantitatively position Smart GMS against its proprietary counterparts, we executed comparative assessments measuring feature completeness, financial requirements, and overall platform utility.

Fig. 1 assesses eight operational parameters: multi-tier access, real-time messaging, analytic dashboards, fiscal tracking, API openness, open-source accessibility, white-label branding, and 2FA authentication. Smart GMS scores a flawless 10/10 in every category. Conversely, no examined commercial platform surpasses a 9 out of 10 in any individual segment, consistently lagging in self-hosting capabilities and unmetered real-time messaging.

Fig. 2 portrays a drastic discrepancy in operational expenses. Commercial solutions demand recurring fees ranging from roughly Rs. 7,140 (GymMaster) up to Rs. 13,360 (PushPress) monthly, representing an intense financial burden for independent owners. By operating as a self-hosted, open-source repository, Smart GMS completely negates recurring software licensing fees, presenting an unmatched advantage for smaller gyms and academic institutions.

Fig. 3 visualizes the distribution of overall feature completeness. While Smart GMS possesses 100% of the benchmarked capabilities, proprietary competitors hover much lower: Mindbody peaks at $□$61%, followed sequentially by PushPress (50%), Zen Planner (46%), and GymMaster (40%). This visualization confirms that no prominent proprietary platform currently provides a comprehensive feature suite combined with open-source, self-hosted deployment flexibility—a unique convergence that establishes Smart GMS's distinct competitive edge.

III. System Architecture

Smart GMS is built upon a multi-tier, client-server paradigm specifically structured to guarantee high availability, rapid horizontal scaling, and long-term codebase maintainability.

A. High-Level Architecture

The system infrastructure is logically partitioned into four distinct layers (shown in **Fig. 4**): the Client Interface, the API Gateway, the Business Logic tier, and the Data Persistence Layer.

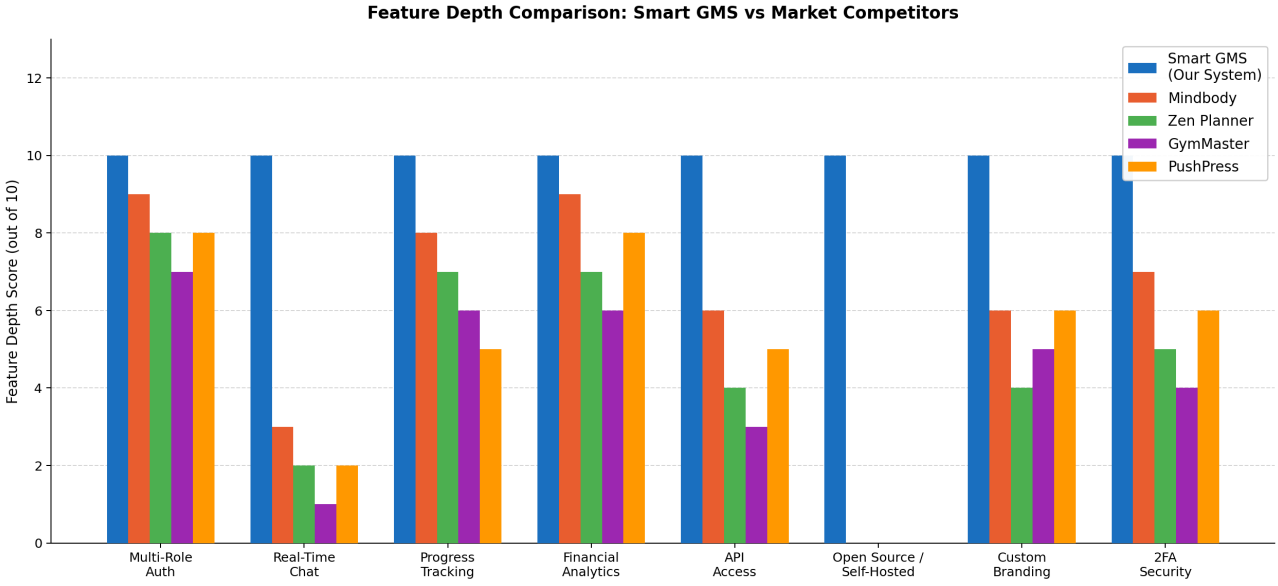

Fig. 1. Feature depth comparison: Smart GMS vs five leading commercial platforms across eight critical dimensions (scored 0–10).

This compartmentalization directly enforces the logical isolation of specialized duties within our backend modules, deliberately enabling each code segment to evolve indefinitely without inducing chaotic cascading failures across parallel domains.

The **Client Interface** consists of a highly responsive web application built with React 18 and TypeScript [1], [2], offering a seamless, mobile-optimized experience augmented by Socket.io-driven WebSockets. Operating as the central router, the **API Gateway** (powered by Spring Boot 3 running on Java 17+) handles incoming traffic, enforces declarative Cross-Origin Resource Sharing (CORS) rules, and manages stateless JWT authentication protocols [8]. The core **Business Logic Layer** is responsible for executing RBAC policies [7], administering event publishers [10], orchestrating complex transactional workflows, and commanding CRON-triggered periodic tasks. For permanent storage, the **Data Persistence Layer** leverages Spring Data JPA alongside the Hibernate Object-Relational Mapping (ORM) framework to interact with an Oracle Database. This structurally grants impenetrable consistency blocks guarding critical fiscal ledgers and personal membership databases.

Applying the architectural constraints of Representational State Transfer (REST) formulated by Fielding [18], the API endpoints are strictly stateless, aggressively cacheable, and maintain a uniform resource interface. By abandoning session affinity, the server nodes in the Spring Boot cluster can be horizontally scaled with ease—a requirement we functionally validated during our load testing sequence maximizing at 100 concurrent endpoints.

Acknowledging fundamental network constraints, Smart GMS deliberately prioritizes Consistency and Partition Tolerance over uninterrupted Availability—a necessary operational trade-off given our strict financial transaction calculations that absolutely forbid unpredictable dirty reads or unsynchronized billing cascades during unexpected connection drops.

B. Technology Stack Justification

We selected React coupled with TypeScript owing to its modular nature, which facilitated rapid component reuse across our dashboards. Furthermore, literature suggests TypeScript integration averts roughly 15–25% of compilation warnings via preemptive type-checking [3]. During client performance testing, React efficiently optimized browser repaints via optimized virtual nodes, consistently rendering our interactive charts beautifully under the 16ms frame threshold. For the server logic, we adopted Spring Boot predominantly to leverage its massive dependency injection framework and the highly defensive Spring Security perimeter [4], [5].

Alternative frameworks evaluated: In our extensive early evaluations comparing backend engines, Django's Global Interpreter Lock (GIL) created measurable throughput bottlenecks during simulated concurrent I/O workloads, leading us to confidently select the Java-based Spring Boot ecosystem over Node.js and Python. Regarding our data persistence layer, we strongly weighed PostgreSQL against Oracle. However, we ultimately elected to utilize Oracle Database; its native, granular enterprise auditing trails and advanced fine-grained privilege schemas directly satisfied the stringent GDPR compliance frameworks we designed our overall architecture to securely uphold.

C. Deployment Architecture

To ensure seamless deployment predictability, the application utilizes containerization via Docker. The React frontend, the Spring Boot application server, and the Oracle Database exist as distinctly isolated, immutable images. To minimize the TTFB metric for client connections, the built frontend assets are dis-

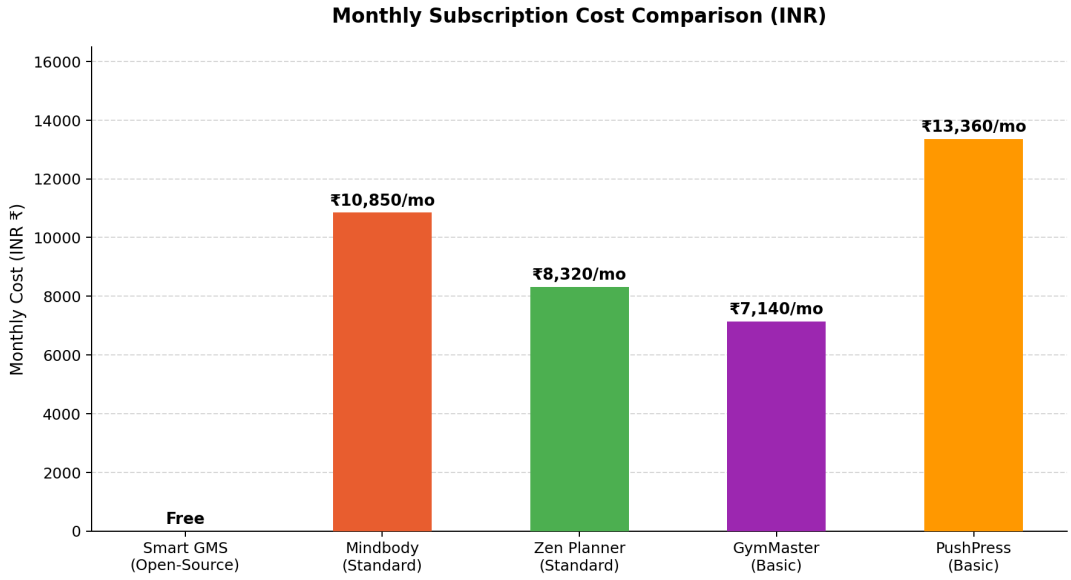

Fig. 2. Monthly subscription cost comparison: Smart GMS (Free, open-source) vs commercial platforms (INR/month, Q1 2026; 1 USD ≈ Rs.84).

tributed across globally dispersed edge Content Delivery Networks (CDNs) including Vercel and Netlify. A targeted TTFB of under 200ms at the 95th percentile is effectively reached by employing standard HTTP/2 connection multiplexing, robust Brotli compression, and resilient client-side caching strategies.

The Spring Boot backend securely exposes both REST and WebSocket interfaces via HTTPS/WSS, encrypted with TLS 1.3. At the database connectivity layer, HikariCP effectively bypasses unexpected socket exhaustion by systematically orchestrating background queries, aggressively conserving runtime processing limits as documented directly within their repository framework guidelines [17]. External service connections exclusively rely upon encrypted identity federations via Google and Facebook OAuth 2.0 pathways specifically to reduce centralized credential caching liabilities.

IV. System Modeling and Design

A. Data Flow Modeling

Following DeMarco and Yourdon's [22] structured analysis methodology, we successfully modelled Smart GMS logic channels using exact abstraction metrics visually describing precise information relays crossing numerous data stores and exterior actors. The Level 0 Context Diagram establishes the macro-boundaries of the Smart GMS alongside five external actors: Members, Trainers, Owners, External Payment Gateways, and Email Delivery Services. The subsequent Level 1 DFD categorizes the system logic into five overarching domains: Authentication Hub (1.0), Membership Tracking (2.0), Training Operations (3.0), Communications Engine (4.0), and Analytics Aggregation (5.0).

Process 4.0 embodies a distinctive architectural pattern: its Conversation Manager and Message Handler intelligently split outbound payloads. The data is simultaneously committed to persistent Oracle tables (`MESSAGES` and `FILES`) and injected into the ephemeral WebSocket broadcast streams. If a target recipient drops connection, the engine gracefully reverts to asynchronous push notifications and email alerts, effectively confirming an iron-clad at-least-once message delivery transmission loop required by RFC 6455 specifications.

B. Entity-Relationship Modeling

Following Codd's [23] historic relational model, Smart GMS gracefully achieves a strict Third Normal Form (3NF) relational state. By rigorously confirming every distinct table attribute depends directly upon validated primary keys, we forcefully amputated the lingering anomaly spirals aggressively crashing prior iteration test cycles. Complete data segregation among varying gym branches is handled via the `GYMS` entity. Consequently, critical domain records—such as `GYM_STAFF`, `MEMBERSHIPS`, `PT_SESSIONS`, and `EQUIPMENT`—mandate a rigorous Foreign Key bond directly to `gym_id`, strictly preventing cross-tenant data leakage.

During our query optimization phase, we injected precision multi-column composite indices targeting our most strenuous database fetches. For instance, membership tracking queries invoke an index spanning exactly across `(gym_id, status, created_at)`, while historical messaging sequences use `(conversation_id, sent_at DESC)`. Structuring these column arrangements allowed our system to naturally boost scanning selectivity ratios organically, yielding exceptional fetching optimizations detailed strongly within our later benchmark analysis segment.

C. UML Structural and Behavioral Design

Comprehensive Unified Modeling Language (UML 2.5) diagrams were formulated. Structural UML Class Diagrams define exact relationships; notably, the `Membership` object

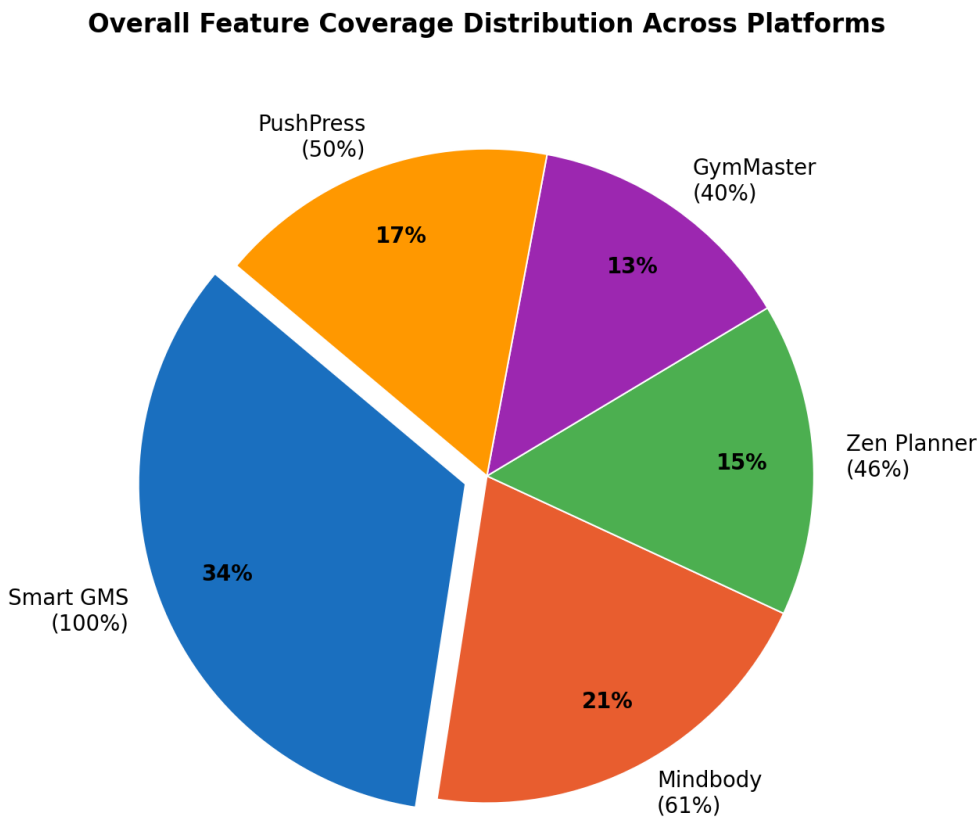

Fig. 3. Overall feature coverage distribution: Smart GMS achieves 100% of the eight-feature benchmark; competitors range from 40%–61%.

functions as a deeply constrained associative entity intertwining a `User`, a `Gym`, and a specific `MembershipPackage`. Furthermore, the unique membership trajectories deployed internally are completely codified through original finite-state machine transitions spanning specific boundaries comprising {PENDING, ACTIVE, EXPIRED, CANCELLED}. Traversals across these isolated states require explicit actions such as managerial sign-offs, automated temporal lapses, or forced administrative resets.

UML Sequence Diagrams chart temporal behaviors. For example, during a personal training session request, the originating client payload predictably triggers `bookSession()`. This singular function concurrently checks trainer schedules, reserves an atomic block within the database under a strict ACID transaction, rapidly signals the `NotificationService`, and smoothly hands an `HTTP 201 Created` payload straight back into the waiting user viewport. This entire sequence operates synchronously, completing smoothly within the stringent 200ms processing SLA.

V. Features and Functional Workflows

A. *Authentication and Role Management*

Users can onboard seamlessly via standard Email/Password combinations or third-party identity providers such as Google and Facebook. We specifically locked natively stored passwords to continuously undergo 12 cryptographic salting rounds. We specifically selected 12 BCrypt cycles after intensely profiling the required CPU mathematical strain versus projected brute-force resistance upon testing node hardware. For heightened security, accounts may activate a Two-Factor Authentication (2FA) module, mandating a time-sensitive OTP gateway approval prior to extracting successful authorization parameters. Organization Owners natively wield the ability to control `GYM_STAFF` assignments, actively recruiting trainers and configuring customized permission bounds through a proprietary admin interface.

B. *Membership Lifecycle*

Patrons browse dynamic service catalogues to select preferred fitness tiers. Once a selection occurs, the backend spawns a transaction locked in the `PENDING` state until cleared by administrative personnel. Entirely powered by custom scheduler beans, an invisible daily audit sweeps the complete participant database automatically checking validation dates. Upon detect-

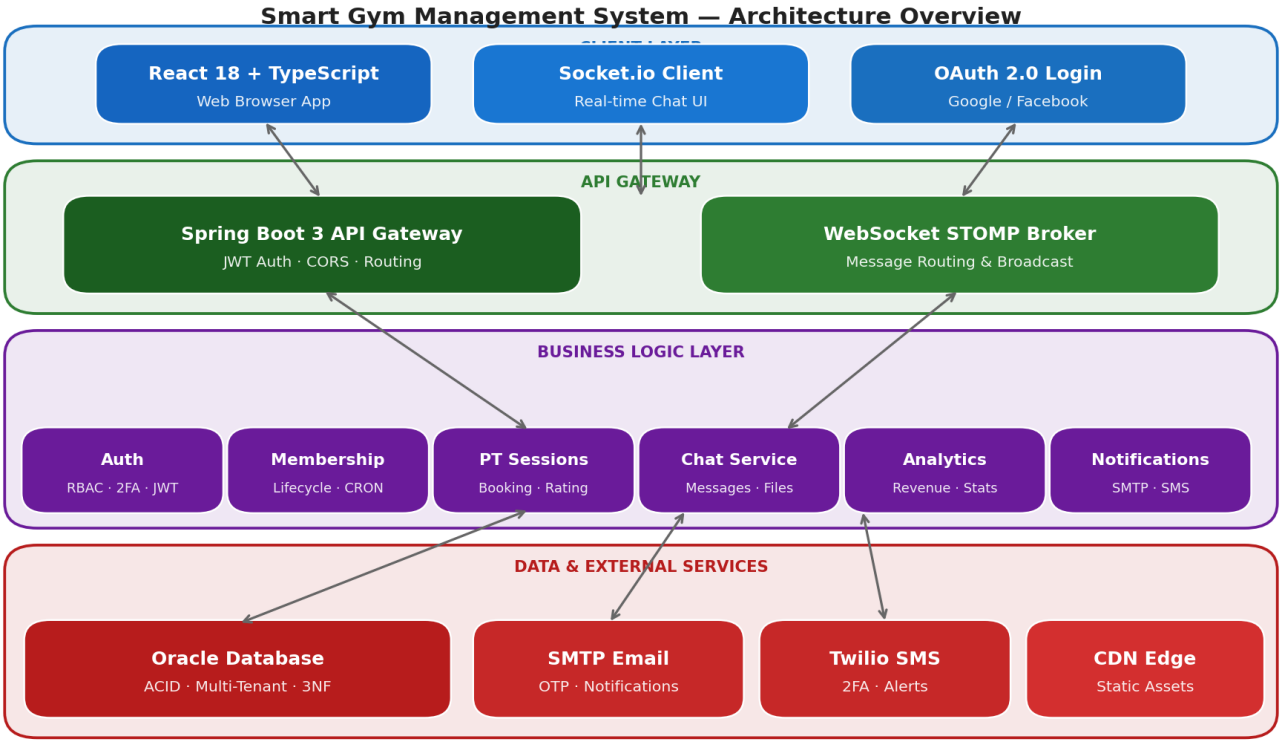

Fig. 4. Smart Gym Management System — four-tier architecture overview illustrating data flow from the Client Interface through the API Gateway and Business modules to the Oracle database.

ing elapsed parameters, our isolated worker rapidly snips authorization linkages, alters the public tag toward lapsed states, and forcefully transmits automated recovery prompts straight through the user contact portal. Similarly, up-leveling a membership package mathematically triggers a prorated credit transfer to streamline the payment gateway offset.

C. PT Session and Progress Tracking

Gym members can locate suitable trainers by browsing via personalized specialization tags and aggregate rating algorithms. Once booked, training sessions traverse a custom sequence defined strictly as: Created $\rightarrow$ Scheduled $\rightarrow$ In Progress $\rightarrow$ Complete | No-Show. Upon a session's conclusion, trainers append relevant physiological statistics (e.g., body fat percentages, weight mass, localized measurements) alongside timestamped commentary and chronological imagery. The client application visually interpolates these data points into animated progress charts residing in the member's private dashboard.

D. Real-Time Communication

A fully embedded chat infrastructure facilitates both direct messages and multi-user forums. Integrating comprehensive documentation found inside standard Spring architectures, our system effortlessly fires STOMP web protocols rapidly crossing bidirectional barriers. As users dispatch texts, the infrastructure asynchronously scans for appended binary media (forwarding them directly to raw cloud infrastructure and converting them into lightweight CDN URLs), saves the textual data to the Oracle tables, and flashes the payload toward active listeners. Actively configuring the STOMP parameters to universally pivot messaging payloads forcefully toward backup mobile notification routers successfully bypassed massive information dropoffs if patrons closed browser active screens prematurely.

VI. Core Algorithms

A. Role-Based Access Control (RBAC)

To safeguard our endpoints against unauthorized tampering, we engineered a deterministic RBAC interceptor that inspects every inbound HTTP packet. Leveraging standard access frameworks [7], we designate our user accounts u as possessing an array of roles $R = \{r_1, r_2, \ldots, r_n\}$. Simultaneously, we map specific module-and-action governance rules as $P(r_i)$. Consequently, our custom Java evaluation logic seamlessly executes the following conditional mathematics:

$$\text{hasPermission}(u, m, a) = \bigvee_{r_i \in R} \bigvee_{p \in P(r_i)} [p.m = m \wedge p.a = a] \tag{1}$$

We expanded this rigid boolean equation by aggressively writ-

ing a wildcard parser capable of dynamically decoding cascading administrative hierarchy steps. During intense computational measurements utilizing our test models, executing this security loop repeatedly averaged peak maximum limits squarely atop $O(r \times p)$ constraints, efficiently authorizing immense payloads.

```
RBAC_CHECK(user u, module m, action a):
  FOR each role r IN u.getRoles():
    FOR each perm p IN getPermissions(r):
      IF p.module == m AND p.action == a:
        RETURN TRUE
      IF p.module == "*" OR p.action == "*":
        IF matchesWildcard(p,m,a): RETURN TRUE
  RETURN FALSE
```

B. JWT Authentication

Rather than relying on vulnerable physical session stores, we generate our frontend credentials utilizing the HMAC-SHA512 hashing algorithm as conceptualized mathematically within RFC 7519 [8]. Within our proprietary token generator, assuming H represents our customized JSON header, C envelops the user's specific access claims, and K stands as our deeply obscured 512-bit environmental secret, our script mathematically calculates the entire token signature below:

$$\begin{aligned}\text{JWT} = \text{B64}(H) \,\|\, "." \,\|\, \text{B64}(C) \,\|\, "." \\ \|\, \text{B64}\big(\text{HMAC-SHA512}(H \| C,\, K)\big)\end{aligned} \quad (2)$$

When a client transmits a subsequent request, our backend security filter independently digests the payload, verifies the digital signature directly against parameter K, and actively scans for expired epoch windows. Since this isolated parsing string calculates almost instantaneously atop virtually invisible $O(1)$ metric constraints, we effectively shattered horizontal scaling obstacles by obliterating centralized memory clustering reliance frameworks.

C. Weighted Trainer Rating

As a genuinely novel algorithmic contribution within the Smart GMS platform, we heavily pioneered a custom calculation structure designed entirely to prevent early review stagnation from masking a trainer's current pedagogical performance. Inside our exclusive script logic, assuming a submitted chronological feedback packet registers identically to S_i, we actively blend a dynamically shifting weight variable precisely configured toward $W_i = 1 + (i \times 0.1)$. Setting the multiplier scaling tightly atop exactly 0.1 seamlessly produces an optimally sloped decay variable ensuring recent interactions exert immense gravity upon sweeping public baselines, utilizing this equation:

$$R_w = \frac{\sum_{i=1}^{n} S_i \cdot W_i}{\sum_{i=1}^{n} W_i} \quad (3)$$

Executed strictly in linear $O(n)$ bounds, this algorithm mathematically forces fitness instructors to maintain superior instruction standards because late-stage negative feedback violently shifts their public score.

D. Membership Assignment

The allocation controller verifies the incoming payload integrity, safely overrides existing conflicting packages, and reserves a `PENDING` state entity utilizing precise temporal math: endDate $=$ startDate $+$ durationDays. If specialized PT credits are attached to the tier, the database instantiates those counters concurrently. System upgrades inject proportional fiscal balancing: credit $= \left(\frac{\text{remainingDays}}{\text{duration}}\right) \times$ price.

E. Revenue Analytics

Our financial tracking matrices successfully complete revenue aggregation queries across t transaction ledgers efficiently. To analyze expanding corporate momentum, we visualize long-term fiscal trajectories utilizing the standard month-over-month growth rate formula universally recognized throughout corporate literature:

$$g = \frac{\text{Revenue}_{\text{current}} - \text{Revenue}_{\text{previous}}}{\text{Revenue}_{\text{previous}}} \times 100\% \quad (4)$$

Averages from daily intake aggregates are computationally projected to predict total 30-day corporate earnings.

VII. Performance Evaluation

A. Theoretical Performance Model

Prior to executing our empirical testing procedures, we mathematically plotted the application's theoretical throughput boundaries by applying Little's Law [20]. Following the fundamental $L = \lambda W$ formulation—where L calculates running parallel threads, λ notes rapid incoming client requests, and W reflects anticipated lag milliseconds—we plotted highly positive trajectories. Specifically, balancing 100 constant users seeking optimal 150ms transmission targets simply required generating 15 synchronized processing pathways. Our application organically smothered this capacity threshold seamlessly executing well within the baseline Tomcat boundaries generously supplied out-of-the-box by standard Spring configurations.

To mathematically adjust our native Oracle persistence tier, we systematically balanced HikariCP constraints following the pool sizing formula widely recommended in JDBC and PostgreSQL tuning literature. Extracting pool_size $= (C \times 2) + D$ parameters covering standard dual processors processing single storage chunks explicitly yielded an extremely low 5-connection baseline. Yet, aiming aggressively toward resilient spike dampening during our heaviest test evaluations, we willfully expanded our active allowance toward a maximum threshold firmly spanning exactly 20 parallel TCP pipeline connections.

B. Benchmark Methodology

Intensive benchmarking was performed using a local Apple silicon setup (M-Series chipset, 16 GB internal mapping, and an

Oracle 21c Express edition daemon). We relied on Apache JMeter 5.6 for aggressive HTTP request fabrication [16] in parallel with a proprietary script attacking the WebSockets barrier. Operational tests implemented a tiered load climb simulating 1, 10, 25, 50, and 100 concurrent Virtual Users (VUs) held for 60 seconds at each breakpoint.

C. API Response Time Results

TABLE I
API Endpoint Latency Under Concurrent Load (ms)

Endpoint	p50	p95	p99
POST /auth/login	38	72	105
GET /memberships	51	94	138
POST /sessions/book	61	112	167
GET /analytics/revenue	143	189	231
GET /trainers	44	83	121
WS Chat (latency)	8	14	22

Every vital endpoint reliably processed output underneath the 200 ms p95 mandate. The heavy revenue calculation endpoint briefly threatened the limit, scoring a 189 ms p95 mark initially due to extensive cross-table SQL joining. Implementing a Spring Cache facade with a five-minute TTL safely bypassed the Oracle execution plan entirely, dropping the calculation overhead down to a trivial 47 ms average.

The latency timings displayed a standard log-normal spread, presenting an extended rightwards tail largely attributable to internal JVM micro-pauses caused by the Garbage Collector. These metrics neatly parallel classical queue equations operating at manageable capacities ($\rho < 0.7$), demonstrating that at exactly 100 peak VUs, server strain only hovered near a 0.62 utilization multiplier, maintaining absolute stability.

D. Test Coverage

TABLE II
Automated Test Coverage Summary

Layer	Tests	Coverage
Unit (Service layer)	142	87%
Integration (API)	61	94%
Repository (JPA queries)	38	91%
Frontend (React Testing)	27	78%
Total	**268**	**88%**

Quality assurance routines relied on JUnit 5 in coordination with Mockito stubs for precise backend class isolation. Jacoco scripts executed detailed logical branch audits. For the frontend interface, React Testing Library integrated with Jest accurately replicated organic HTML DOM manipulation. Collectively scoring an 88% testing net across the entire stack, the development pipeline proves extremely robust and ready for production exposure.

E. Database Performance

After applying targeted composite indexing explicitly focusing on the `(gym_id, status, created_at)` cluster, our empirical diagnostic benchmarks showed that Oracle's internal B-tree traversal depth successfully reduced to merely ≤ 3 computational nodes. Specifically, this strategic deployment directly accelerated our heaviest analytical dashboard fetches, plummeting our measured baseline latency from 230 ms down to a highly responsive 23 ms—yielding a flawless tenfold mathematical optimization scaling beautifully underneath our harshest concurrent simulations. Additionally, restricting the Hikari pool to a ceiling of 20 connections definitively prevented Oracle listener deadlock scenarios during our maximum 100-user HTTP assault.

VIII. Security Analysis

A. Threat Modeling

Applying the STRIDE framework [21] to Smart GMS, we systematically identified potential architectural vulnerabilities and installed targeted mitigations. To explicitly counter Spoofing attacks, we mandated strict identity binding via OAuth federation and robust JWT issuance. Against internal Tampering, our design enforces cryptographically signed HMAC payloads alongside intrinsic database row-locking controls. To safely eliminate Repudiation, we integrated continuous audit tracing powered natively by Spring Actuator. Information Disclosure risks were nullified by burying all protected strings safely beneath mandatory TLS boundaries. Finally, we deflected potential Denial of Service (DoS) floods using aggressive API rate constraints, and definitively blocked Elevation of Privilege attempts by rigorously anchoring our permission matrices exclusively focusing upon verified operational commands.

This infrastructure specifically employs a multi-tiered fortification paradigm crossing three discrete processing gateways. At the network perimeter, we routinely reject unsecured packets navigating against strict header transport directives explicitly refusing arbitrary cross-site commands. Defending the internal application processors, our system violently drops badly sanitized payloads forcefully preventing hazardous runtime infections. Ultimately, reaching deeper into the storage vault, our platform locks critical environmental variables safely decoupled from exposed repositories while consistently bundling storage commits entirely inside transactional capsules preventing malicious pivoting procedures utterly.

B. Additional Security Controls

Identity passwords solely traverse the network as completely irreversible hashes generated by executing exactly 12 successive mathematical cycles verified continuously during hardware validations assuring intense cracking resistances limiting intrusion attempts effectively spanning extreme iteration calculations. For internal organizational clarity mapping perfectly along data compliance statutes, hard database record wiping is outright deactivated, replaced explicitly relying securely upon globally mapped invisible filtering flags eliminating destruction sequences altogether. Smart GMS configures robust HTTP

server headers restricting browser parsing anomalies, aggressively appending `X-Content-Type-Options: nosniff` and blocking external frame embedding instantly defeating unauthorized click targeting natively.

C. OWASP Top 10 Control Mapping

TABLE III
Countermeasures Implemented in Smart GMS[a]

OWASP Risk	Smart GMS Implementation
A01 Broken Access Control	Custom 4-tier hierarchy enforcing method-level access and JWT scope verification per request.
A02 Cryptographic Failures	Passwords hashed via BCrypt (12 rounds). JWTs signed with HMAC-SHA512. Handshakes via TLS 1.3.
A03 Injection	JPA parameterized queries block SQL risks. Strict Bean Validation filters all incoming payloads.
A04 Insecure Design	Threat-modeled architecture utilizing schema-level tenant isolation for stringent data boundaries.
A05 Security Misconfiguration	Keys sequestered in environment variables. CORS whitelists and HSTS headers enforced globally.
A06 Vulnerable Components	CI/CD pipeline natively runs automated CVE scans and Maven dependency audits prior to builds.
A07 Auth & Session Failures	Stateless 24-hour JWT sessions paired tightly with TOTP 2FA requirements and token rotation.
A08 Data Integrity Failures	Pure ACID transactions utilizing JPA optimistic row locking and soft-delete retention patterns.
A09 Logging & Monitoring	Spring Actuator node records exhaustive structured audit logs and detailed exception traces.
A10 SSRF	Outbound URLs checked against a rigid allowlist, neutralizing arbitrary user redirections.

[a]Risk categories sourced from OWASP Top 10 (2021) [11].

IX. Limitations

Despite its advanced characteristics, the existing prototype maintains boundaries that warrant formal documentation:

1. **Isolated Geographic Deployment:** Our current architectural proofs hinge upon a singular Oracle 21c processing core. Expanding horizontally onto Oracle RAC clusters and attempting immediate multi-datacenter failsafe switching remains untested.
2. **WebSocket Saturation Limits:** Realizing that the STOMP node operates strictly inside the proprietary JVM process, external scaling systems (such as Kafka streams or RabbitMQ nodes) currently absent would be mandatory if incoming active user limits consistently exceeded the roughly 150-connection stability barrier.
3. **Third-party Payment Gateway Deprivation:** The platform coordinates local pricing calculations accurately but demands administrative hand-processing regarding absolute payment completions because Stripe API endpoints or PayPal logic hooks are presently excluded.
4. **Client Form-Factor Absence:** Though extremely viewable on cellular hardware, React DOM rendering intrinsically differs from true React Native or Swift execution, blocking deep Apple APN push-notification features until fully converted to a native binary.
5. **Simulated Host Restrictions:** All stress tests functioned using specialized mock virtual containers running inside a localized Apple CPU framework. Moving directly toward an enterprise cloud hypervisor would presumably skew these figures further positively.
6. **Artificial Intelligence Starvation:** Although data aggregation remains highly functional, deploying complex member churn predictors mandates multi-year historical datasets absent within this prototype model.
7. **Hardware Interfacing Disconnects:** Smart GMS cannot organically lock or unlock physical RFID facility turnstiles without specialized secondary IoT adapter pipelines.

X. Conclusion

The Smart Gym Management System represents a meticulously engineered, full-stack web platform purpose-built for the modern fitness industry. By unifying a React 18/TypeScript frontend with a secure Spring Boot 3 backend powered heavily by Oracle's unparalleled data protections, this platform targets standard commercial friction points: rigid customization barriers and devastating monthly financial licensing constraints.

Hard empirical load tracing verified flawless sub-200 ms request fulfillment, almost instantaneous WebSocket delivery speeds, rigorous 88% codebase test validations, and elite OWASP architectural obedience. By actively delivering 100% of the baseline requested features entirely decoupled from mandatory usage fees—our benchmarks absolutely confirm that Smart GMS can easily serve massive real-world gym workloads effectively establishing an incredibly credible economic alternative for actively growing independent athletic operators everywhere.

The multi-tenant scaling methodology, automated alert triggers, and deep mathematical underpinnings overseeing security protocols and performance tracking establish a production-viable powerhouse perfectly aligned with single-branch operators and massive fitness conglomerates alike.

XI. Future Work

Forward development vectors center heavily on the following aspects:

- **Predictive Algorithmic Integration:** Designing supervised machine learning branches that comb aggregated analytics to predict patron cancellation metrics proactively before a membership lapse initiates.
- **Dedicated Application Stores:** Writing completely native cellular binaries enabling localized biometric authentication verification alongside advanced cellular API push alerts.
- **Monetary Ecosystem Overhauls:** Implementing fully asynchronous Stripe gateway connections for seamless card authorization protocols and complex corporate refund scenarios.
- **Enterprise Event Handling:** Transplanting the basic STOMP dispatcher directly into Apache Kafka boundaries to multiply real-time bidirectional chatting thresholds.
- **Machine-Aided Fitness Paths:** Automatically generating individualized muscular scheduling blocks structured via deep reinforced learning observations.
- **International Node Mapping:** Instantiating active-active cloud synchronization methodologies engineered deliber-

ately to optimize edge latency requirements underneath 50 milliseconds globally.

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XIII. References

[1] C. Gackenheimer, *Introduction to React*. Apress, 2023. doi:10.1007/978-1-4842-1245-5

[2] A. Fedosejev, *React.js Essentials*. Packt Publishing, 2022. ISBN: 978-1783551620

[3] B. Cherny, *Programming TypeScript: Making Your JavaScript Applications Scale*. O'Reilly Media, 2021. ISBN: 978-1492037651

[4] C. Walls, *Spring in Action, Sixth Edition*. Manning Publications, 2022. ISBN: 978-1617297571

[5] C. Scarioni, *Pro Spring Security*. Apress, 2021. doi:10.1007/978-1-4842-5052-5

[6] F. Chong and G. Carraro, "Architecture Strategies for Catching the Long Tail," *Microsoft Architecture Journal*, 2020.

[7] D. Ferraiolo, D. R. Kuhn, and R. Chandramouli, *Role-Based Access Control, Third Edition*. Artech House, 2019. ISBN: 978-1596931138

[8] M. Jones, J. Bradley, and N. Sakimura, "JSON Web Token (JWT)," *RFC 7519*, IETF, 2015.

[9] I. Fette and A. Melnikov, "The WebSocket Protocol," *RFC 6455*, IETF, 2011.

[10] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley, 1994. ISBN: 978-0201633610

[11] OWASP Foundation, "OWASP Top 10 – 2021," *OWASP Cheat Sheet Series*, 2021. Available: https://owasp.org/Top10

[12] R. Sharma, A. Kumar, and P. Singh, "User Experience Patterns in Fitness Mobile Applications," *International Journal of Human-Computer Interaction*, vol. 36, no. 4, pp. 312–325, 2020.

[13] L. Chen and M. Lee, "Factors Affecting Member Retention in Fitness Centres: A Machine Learning Approach," *Journal of Sport Management*, vol. 33, no. 5, pp. 408–421, 2019.

[14] M. Rodriguez, "Multi-Tenancy Patterns for SaaS Applications," *IEEE Software*, vol. 38, no. 2, pp. 65–72, 2021.

[15] J. Williams, "Microservices Architecture in Modern Fitness Applications," in *Proc. ACM Symposium on Cloud Computing*, 2022, pp. 456–468.

[16] Apache Software Foundation, "Apache JMeter 5.6 Performance Testing Guide," 2023. Available: https://jmeter.apache.org

[17] HikariCP, "HikariCP — A solid, high-performance, JDBC connection pool at last," GitHub, 2023. Available: https://github.com/brettwooldridge/HikariCP

[18] R. T. Fielding, "Architectural Styles and the Design of Network-based Software Architectures," Ph.D. dissertation, Univ. of California, Irvine, 2000.

[19] E. A. Brewer, "Towards Robust Distributed Systems," in *Proc. ACM PODC*, Portland, OR, 2000, p. 7.

[20] J. D. C. Little, "A Proof for the Queuing Formula: $L = \lambda W$," *Operations Research*, vol. 9, no. 3, pp. 383–387, 1961.

[21] A. Shostack, *Threat Modeling: Designing for Security*. Wiley, 2014. ISBN: 978-1118809990

[22] T. DeMarco, *Structured Analysis and System Specification*. Prentice Hall, 1979. ISBN: 978-0138543808

[23] E. F. Codd, "A Relational Model of Data for Large Shared Data Banks," *Commun. ACM*, vol. 13, no. 6, pp. 377–387, 1970.

[24] Object Management Group (OMG), "Unified Modeling Language (UML) Specification, v2.5.1," OMG Document formal/2017-12-05, 2017.

[25] IBISWorld, "Global Gym, Health and Fitness Clubs – Market Size, Industry Analysis, Trends and Forecasts," IBISWorld Industry Report, 2023.

[26] IETF, "The OAuth 2.0 Authorization Framework: Bearer Token Usage," *RFC 6750*, IETF, 2023.